

ENVS 193DS homework 03

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Table of contents

Set up	2
Problems	2
Problem 1. Giant kelp fronds	2
a. An appropriate test	2
b. Create a visualization	2
c. Check your assumptions and run your test	3
Checking Assumptions	3
Running Correlation test	4
d. Results communication	5
e. Test implications	5
f. Double check your own work	5
Problem 2. Personal data	6
a. Updating your visualizations	6
Figure 1: Categorical Variable	6
Figure 2: Continuous variable	7
b. Captions	8
Problem 3. Affective visualization	9
a. Describe in words what an affective visualization could look like for your personal data	9
b. Create a sketch (on paper) of your idea	9
c. Make a draft of your visualization	11
d. Write an artist statement	12
e. Prep your materials to share in class	12
Problem 4. Statistical critique	12
a. Revisit and summarize	12

b. Visual clarity	13
c. Aesthetic clarity	13
d. Recommendations	13

Set up

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)

kelp <- read_csv(here("data", "temp-kelp.csv"))
my_data <- read_xlsx(here("data", "my_data.xlsx"))
```

Problems

Problem 1. Giant kelp fronds

a. An appropriate test

An appropriate test to determine the strength of the relationship between temperature and giant kelp frond elongation rate would be to measure the correlation. You can measure strength of correlation through the following tests: Pearson's product-moment correlation, Spearman's rank correlation. Pearson's correlation is parametric and describes linear relationships between variables where variables are continuous and normally distributed, and observations are independent. Meanwhile, Spearman's is non-parametric and describes a monotonic relationship between variables and independent observations.

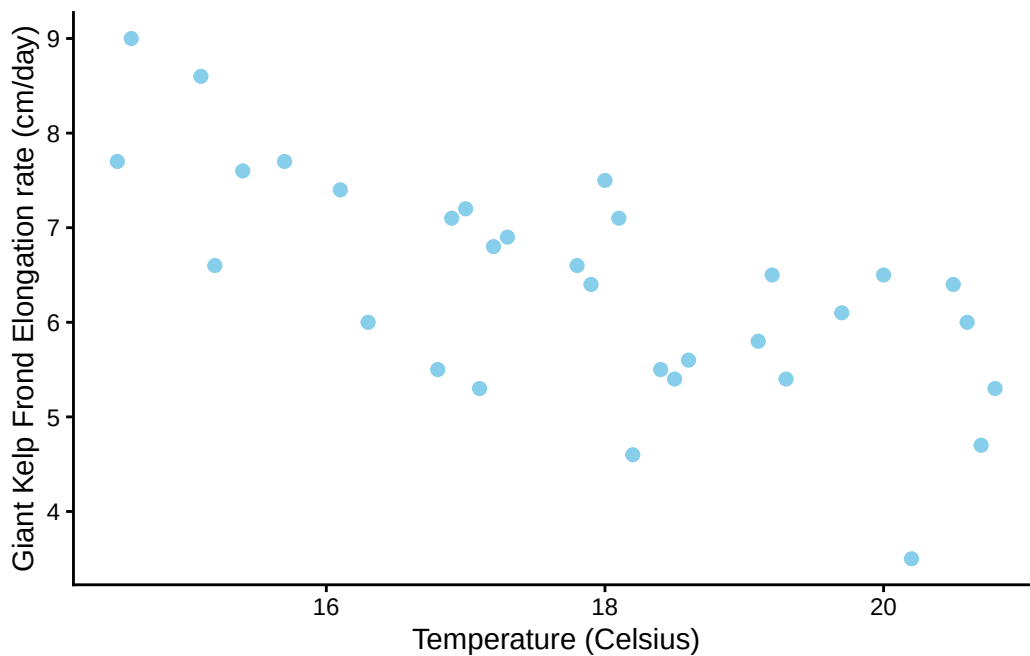
b. Create a visualization

```
# base layer ggplot
ggplot(data = kelp,
       # x axis = temperature
       # y axis = kelp_elong
       mapping = aes( x = temp_c,
                      y = kelp_elong))
```

```

) +
# adding geometry - points, scatterplot
geom_point(size = 2,
           # changing color from default
           color = "skyblue") +
# changing theme from default
theme_classic() +
# adding axes labels
labs(x = "Temperature (Celsius)",
     y = "Giant Kelp Frond Elongation rate (cm/day)")

```



c. Check your assumptions and run your test

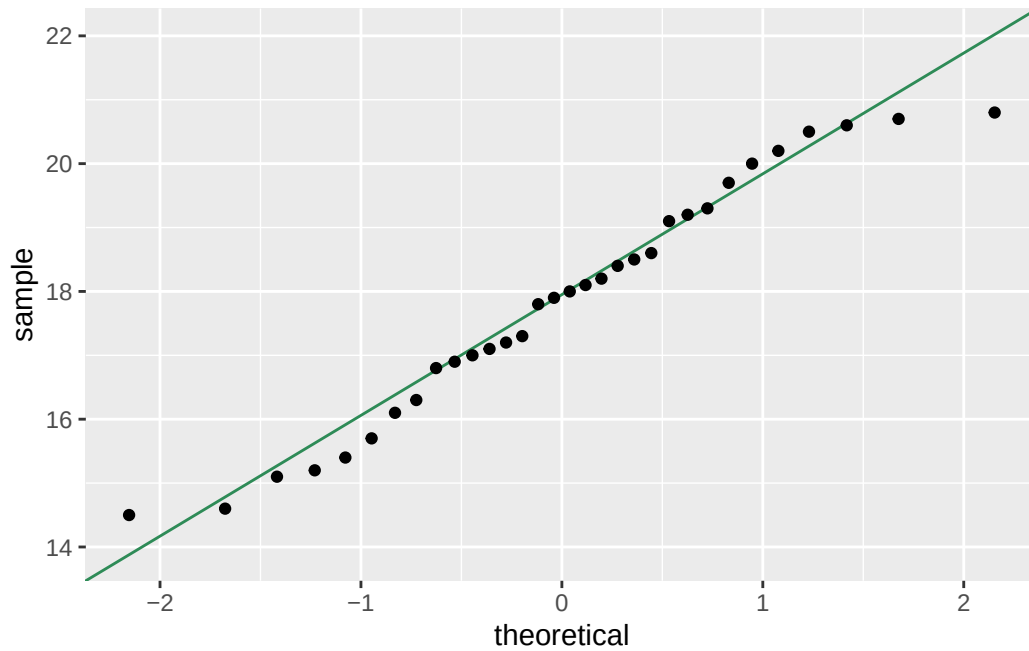
Checking Assumptions

```

# QQ plot to check assumptions
#base layer: urchins_clean
ggplot(data = kelp,
       #y axis: temp_c
       mapping = aes(sample = temp_c)) +

```

```
#first layer: reference for qq plot
geom_qq_line(color = "seagreen") +
#second layer: points representing sample quantiles
geom_qq()
```



For assumptions I wanted to check the assumptions: variables are continuous and normally distributed, and observations are independent using a QQ plot. QQ plots demonstrate normal distribution. The results of which indicated a normal distribution as points lined up with the QQ line.

Running Correlation test

```
# running pearsons coefficient
# data$predictor, data$response
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "pearson")
```

Pearson's product-moment correlation

```

data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489

```

d. Results communication

I used a Pearson's Correlation test in order to determine the strength of the relationship between temperature and giant kelp frond elongation rate. I found a moderate negative relationship between temperature and giant kelp frond elongation rate (Pearson's $r = -0.69$, $t(30) = -5.2$, $p < 0.001$, $\alpha = 0.05$).

e. Test implications

We found a moderate negative (Pearson's $r = -0.69$) relationship between temperature and giant kelp frond elongation rate. Here, as temperature increases, giant kelp frond elongation rate decreases. Therefore, giant kelp frond health increases at lower temperatures and lower temperatures are important for kelp health.

f. Double check your own work

```

# double checking using Spearmans
cor.test(kelp$temp_c, kelp$kelp_elong,
         method = "spearman")

```

Spearman's rank correlation rho

```

data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684

```

After running the Spearman's rank test as well, I have concluded that the results lead me to the same conclusion of rejecting the null hypothesis for a moderate negative relationship between temperature and kelp frond elongation rate. Both tests have similar correlation values -0.68, along with p-values under 0.001 (Spearman $\rho = -0.69$, $S = 9216.1$, $p < 0.001$, $\alpha = 0.05$).

Problem 2. Personal data

a. Updating your visualizations

Figure 1: Categorical Variable

```
# cleaning my data
# creating new object my_data_clean
my_data_clean <- clean_names(my_data) |>
  #making readable
  mutate(work_out = case_when(
    work_out == "N" ~ "No",
    work_out == "Y" ~ "Yes"
  ))

#base layer: ggplot
ggplot(data = my_data_clean,
  #x- axis: did I work out the day before?
  #y- axis: ounces of coffee consumed per day
  mapping = aes(x = work_out,
    y = coffee_oz,
    #grouping colors by work out
    color = work_out)) +

  #adding points
  geom_point(size = 2) +
  #labelling axes
  labs(x = "Work Out the Day Before?",
    y = "Coffee Per Day (oz)",
    title = str_wrap("Working out the day before influences amount of coffee consumed",
      width = 40),
    subtitle = "5/27/26") +
  #choosing different colors, yes = green, no = red
  scale_color_manual(values = c("Yes" = "forestgreen",
    "No" = "tomato")) +
  #changing theme from default and to make colors more visible
```

```
theme_classic() +
#getting rid of legend
theme(legend.position = "none")
```

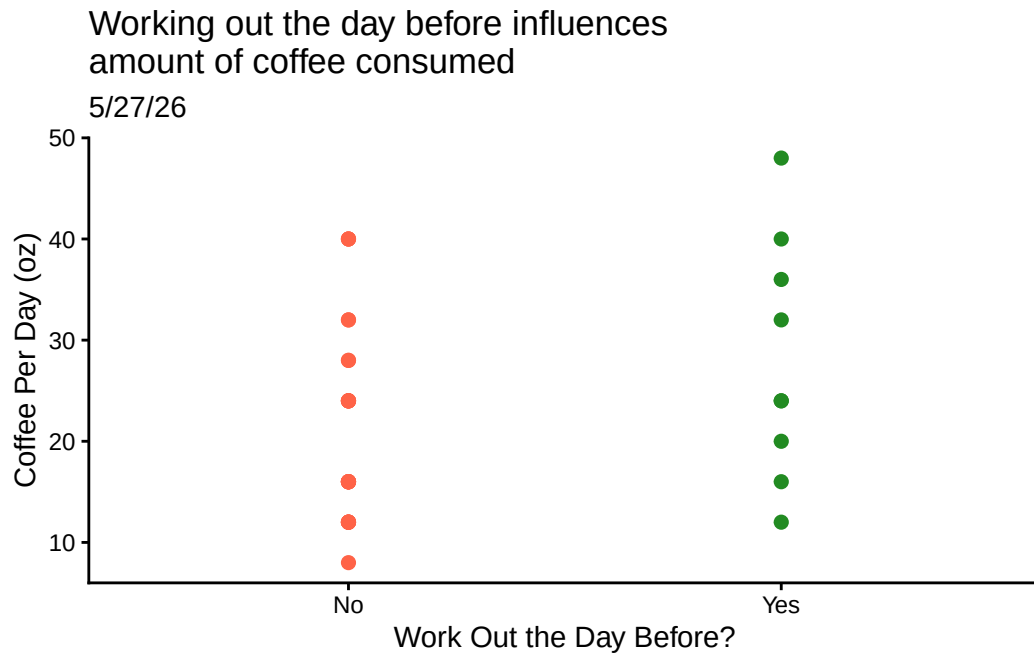


Figure 2: Continuous variable

```
#base layer: ggplot
ggplot(data = my_data_clean,
  #x- axis: Amount of sleep (hours)
  #y- axis: ounces of coffee consumed per day
  mapping = aes(x = amount_of_sleep,
    y = coffee_oz
  )) +

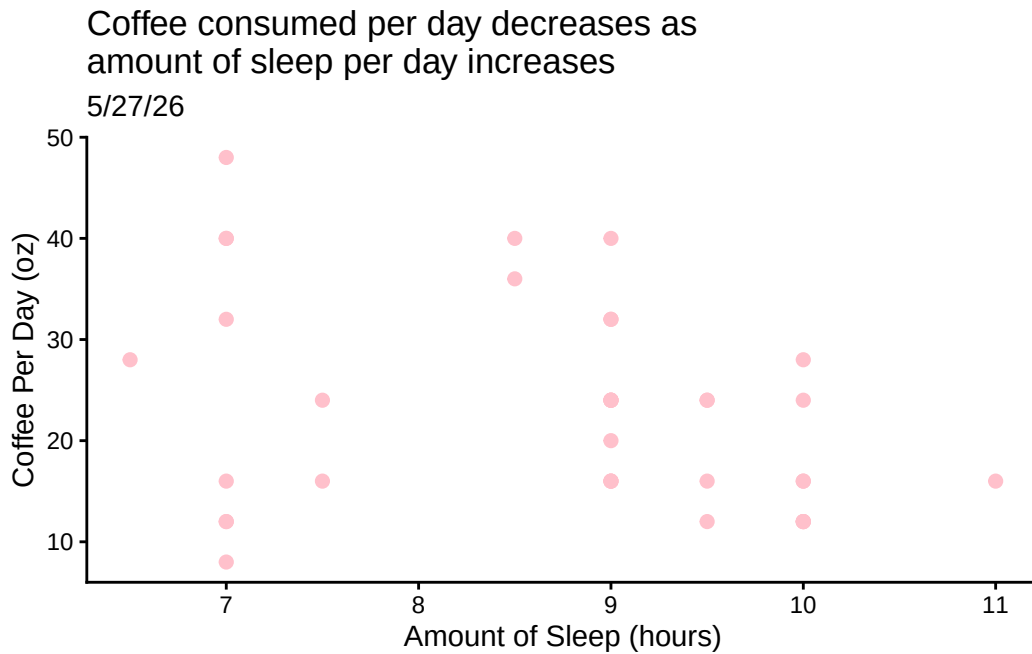
#adding points
geom_point(size = 2,
  color = "pink") +

#labelling axes
labs(x = "Amount of Sleep (hours)",
  y = "Coffee Per Day (oz)",
  title = str_wrap("Coffee consumed per day decreases as amount of sleep per day increases"))
```

```

width = 40),
  subtitle = "5/27/26") +
#changing theme from default
theme_classic() +
#getting rid of legend
theme(legend.position = "none")

```



b. Captions

Figure 1: Working out the day before has little impact on coffee consumed per day (oz). Points represent observations of how much coffee, in ounces, was consumed depending on if I worked out the day before or not (total $n = 37$). Colors represent if I worked out the day before or not (green: Yes, red: No)

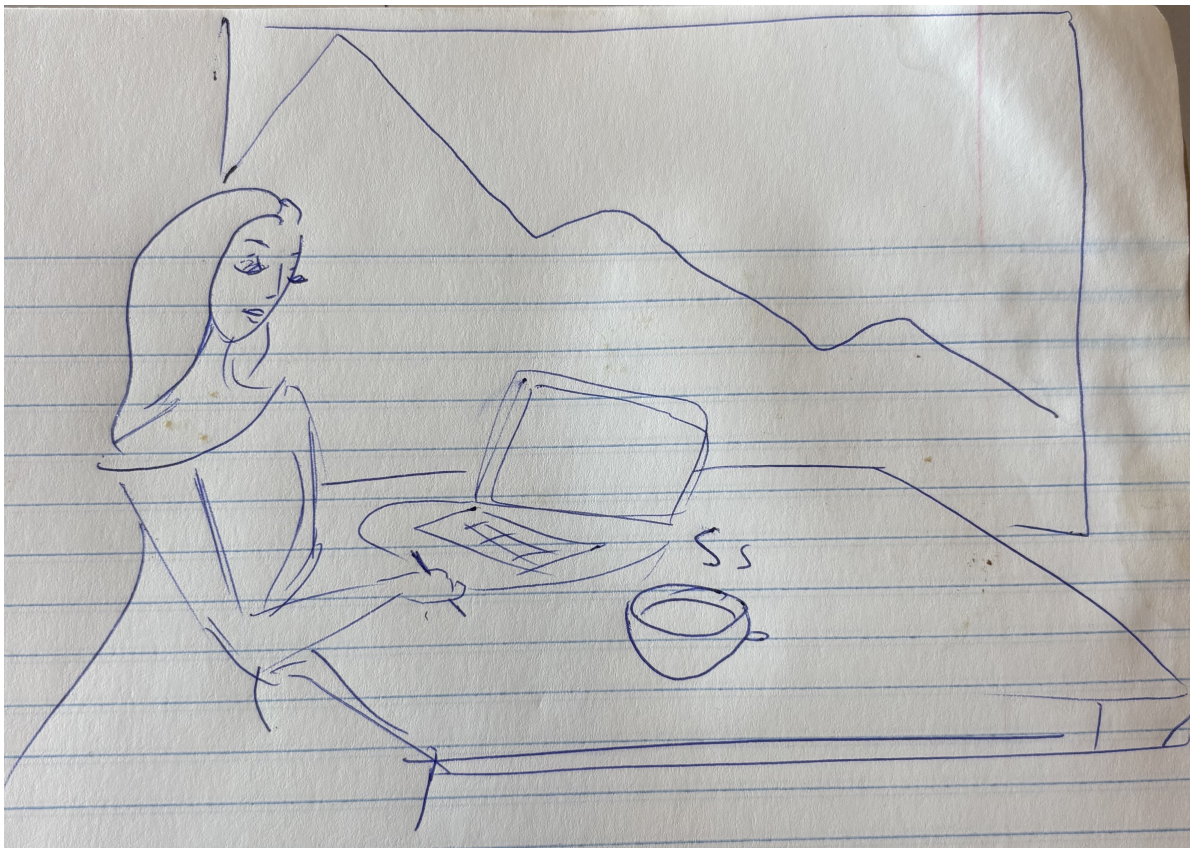
Figure 2: Coffee per day (oz) decreases as sleep (hrs) increases. Points represent observations of how much coffee, in ounces, was consumed compared to how many hours I had slept ($n = 37$).

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

Because I like drinking coffee, I thought it would be nice to recreate my favorite way to drink coffee. Which is at a cafe or at home, cozy inside, with a great view of nature, mountains or otherwise, while I do my work. I was inspired from the watercolor mountain data, and I think it would be visually appealing to show my data as a mountain slope in the background of this coffee scene.

b. Create a sketch (on paper) of your idea



c. Make a draft of your visualization



d. Write an artist statement

My visualization shows the data of coffee consumed per day compared to hours of sleep, in the backdrop of my favorite study setting, drinking a coffee in front of a natural landscape. I really like a simple art style as well as the watercolor landscapes that we were shown, so I wanted to incorporate both into the sketch of me drinking coffee and also the mountains in the background representing the data. I love procreate so I used procreate to make my drawings, but I also wanted to incorporate the watercolor aesthetic so I incorporated the watercolor aspect as well. I created this by inputting a screenshot of my data to trace the mountains onto, then using different layers on procreate to add the water color, the top layer of my drawing, and the mountain landscape.

e. Prep your materials to share in class

[View](#)

Problem 4. Statistical critique

a. Revisit and summarize

The paper uses a Welch's ANOVA test with values and preferences distances as the response variable and effect of gender, age, and income as the predictor variable.

Table 4. Differences of values and preferences distances to respondents' home between clusters and neighbourhoods (Welch's ANOVA).

	Values distance	Preferences distance
Between clusters	F (3,218.3)=0.741,p=0.529	F (3,134.73)=0.834, p=0.476
Between neighbourhoods	F (4,432.32)=17.730, p<0.001***	F (4,197.15)=10.152, p<0.001***

Notes: *p<0.1; **p<0.05; ***p<0.01.

b. Visual clarity

I inserted a table for part a, and it displays the results fairly clearly for an ANOVA test, you are able to see which values belong where. However, I was a little unclear to what the values actually meant and it took a lot of back and forth with reading the text to see what exactly they were talking about.

c. Aesthetic clarity

There was not a lot of “visual clutter”. Titles were in bold and easy to read, however, nothing else was italicized or in bold. Overall, it was straightforward, simple, and easy to read.

d. Recommendations

It was a little difficult to understand so if I had to make recommendations I would say to change the titles to be more descriptive. Although the brevity means less ink, I think the titles are a little too concise so they get a little confusing. Additionally the titles “between clusters” and “between neighborhoods” are a little misleading because it seems like you are measuring the difference between clusters or between neighborhoods instead of between a cluster and a respondents home or between a neighborhood and a respondents' home.